Scheduling of irrigation for paddy

Short duration variety			Medium duration variety			Long duration variety		
Days	No. of irrigation	Water level (cm)	Days	No. of irrigation	Water level (cm)	Days	No. of irrigation	Water level (cm)
1-25	5-7	2-3	1-30	5-7	2-3	1-35	6-8	2-3
25	-	Thin film of water	30	-	Thin film of water	35	-	Thin film of water
28	-	Life irrigation	33	-	Life irrigation	38	-	Life irrigation
29-50	6	2-5	34-65	6-8	2-5	39-90 or 95	12-15	2-5
51-70	5-6	2-5	66-95	8-10	2-5	96-125	7-9	2-5
71-105	5-6	2-5	96-125	6-8	2-5	126-150	5-6	2-5

Water Management for Different Ecosystem

Irrigate after crackline formation

System of Rice Intensification (SRI)

- Irrigation only to moist the soil in the early period of 10 days
- Restoring irrigation to a maximum depth of 2.5cm after development of hairline cracks in the soil until panicle initiation
- Increasing irrigation depth to 5.0 cm after PI one day after disappearance of ponded water.

Transplanted lowland rice

- Puddling and leveling minimizes the water requirement
- Plough with tractor drawn cage wheel to reduce percolation losses and to save water requirement up to 20%.
- Maintain 2.5 cm water over the puddle and allow the green manure to decompose for a minimum of 7 days in the case of less fibrous plants like sun hemp and 15 days for more fibrous green manure plants like Kolinchi (*Tephrosia purpurea*).
- At the time of transplanting, a shallow depth of 2 cm of water is adequate since high depth of water will lead to deep planting resulting in reduction of tillering.
- Maintain 2 cm of water up to seven days of transplanting.
- After the establishment stage, cyclic submergence of water is the best practice for rice crop. This cyclic 5 cm submergence has to be continued throughout the crop period.

Cyclic submergence of 5 cm water

Drain excess water

Wet Seeded Puddled Lowland Rice

- During first one week just wet the soil by thin film of water.
- Depth of irrigation may be increased to 2.5cm progressively along the crop age.
- Provide adequate drainage facilities to drain excess water or strictly follow irrigation schedule of one day after disappearance of ponded water. Last irrigation may be 15 days ahead of harvest.

Perfect levelling for efficient water management

Dry Seeded Un-Puddled Rice

- As that of irrigated rice, canal water is used for irrigation
- For drill sown rice care should be taken to drain out excess rain water during first 10-15 DAS and the water level in the field should not be more than 2.5cm height during tillering stage.
- In drill sown rice carry out hodta operation (Planking) in standing water at 40 DAS, and impound sufficient rain water through the crop growth period.

Crop is irrigated using canal water

Take care to avoid development of cracks

Precautions for irrigation (common)

- The field plot can be 25 to 50 cents depending on the source of irrigation.
- Field to field irrigation should be avoided.
- Small bund may be formed parallel to the main bund of the field at a distance of 30 to 45cm within the field to avoid leakages of water through main bund crevices.
- To minimize percolation loss, the depth of stagnated water should be 5cm or less.
- In water logged condition, form open drains, about 60cm in depth and 45cm width across the field.
- Care should be taken not to allow development of cracks.
- In double cropped wetland of command area, raise groundnut / pulse in the place of *Kuruvai* rice if water is a constraint.

Avoid improper levelling

Proper drainage removes excess water

Excess water and crop performance

- Response of rice crop to excess water conditions is different for different growth stages, varieties and seasons.
- Turbid water greatly damages the crop than clear water.
- Seedling establishment to maximum tillering stage is least sensitive. Excess water during the vegetative phase hinders tillering.
- Flowering to maturity is more sensitive to excess water than the stage from maximum tillering to flowering.
- Fields that are kept flooded beyond the dough stage will mature in a non-homogenous way and harvest will be delayed.

Management of Excess Water

- Levelling of land removes excess water by runoff.
- Drainage removes excess water from the root zone that is harmful for plant growth.
- The best time for drainage in ill drained soils is at late tillering stage.
- Controlled irrigation reduces excess use of water in water-logged area.
- Flood control measures: Construction of bunds may check water flow from the rivers to the cultivable lands.